Complete

JAVA BACKEND DEVELOPER ROADMAP FOR 2027



Java Core

01

Syntax, logic and the JVM until it feels automatic.

WHAT TO LEARN

- JDK, JRE and JVM
- Operators
- Loops
- Arrays
- Wrapper classes
- Variables and data types
- if, else and switch
- Methods
- Strings and StringBuilder
- Scanner and console I/O

TIME

3 weeks

BUILD IT

A console ATM with PIN and balance checks

INTERVIEWERS ASK

Why are Strings immutable? == vs equals()

Object-Oriented Programming

02

Model the real world in classes, not scripts.

WHAT TO LEARN

- Classes and objects
- Encapsulation
- Polymorphism
- Interfaces
- static and final
- Constructors
- Inheritance
- Abstraction
- Access modifiers
- this and super

TIME

2 weeks

BUILD IT

A library system with books, members and fines

INTERVIEWERS ASK

Abstract class vs interface. Overloading vs overriding

Collections and Exceptions

03

Store data smartly. Fail gracefully.

WHAT TO LEARN

- List and ArrayList
- Map and HashMap
- Generics
- Iterator
- Checked vs unchecked
- Set and HashSet
- Queue and Deque
- Comparable, Comparator
- try, catch, finally
- Custom exceptions

TIME

2 weeks

BUILD IT

A contact book with search, sort and bad-input handling

INTERVIEWERS ASK

How does HashMap work internally? ArrayList vs LinkedList

Modern Java and Concurrency

04

Write less code. Run more of it at once.

WHAT TO LEARN

- Lambdas
- Streams API
- Method references
- Threads and lifecycle
- synchronized, volatile
- Functional interfaces
- Optional
- Date and Time API
- ExecutorService
- CompletableFuture

TIME

3 weeks

BUILD IT

A multi-threaded word counter using streams

INTERVIEWERS ASK

map vs flatMap. What causes a deadlock and how to fix it?

Data Structures and Algorithms

05

The currency of every coding round. Run it alongside the other stops.

WHAT TO LEARN

- Big-O analysis
- Linked lists
- Hashing
- Trees and BSTs
- Sorting and searching
- Arrays and strings
- Stacks and queues
- Recursion
- Heaps and graphs
- Dynamic programming

TIME

Daily

BUILD IT

A habit: 1 or 2 problems every day, tracked in a sheet

INTERVIEWERS ASK

Two-pointer and sliding window problems, then explain your Big-O

SQL and Databases

06

Every backend talks to a database. Learn its language.

WHAT TO LEARN

- CRUD queries
- GROUP BY, aggregates
- Indexes
- ACID properties
- PostgreSQL or MySQL
- Joins
- Subqueries
- Transactions
- Normalization
- NoSQL basics

TIME

2 weeks

BUILD IT

An e-commerce schema with 6 tables and 15 queries

INTERVIEWERS ASK

Find the 2nd highest salary. Clustered vs non-clustered index

Git and Dev Tools

07

Work the way real teams work.

WHAT TO LEARN

- Commit, push, pull
- Merge vs rebase
- Pull requests
- Maven and Gradle
- Linux terminal basics
- Branching
- Merge conflicts
- GitHub workflow
- IntelliJ debugging
- Writing a good README

TIME

1 week

BUILD IT

Push every project so far to GitHub, each with a README

INTERVIEWERS ASK

Merge vs rebase. How do you resolve a conflict?

Spring Boot and REST APIs

08

Where you start working as a backend developer.

WHAT TO LEARN

- IoC and DI
- REST and HTTP verbs
- Bean validation
- Profiles and config
- Postman testing
- Spring Boot starters
- Controllers and DTOs
- Global error handling
- OpenAPI and Swagger
- JUnit and Mockito

TIME

4 weeks

BUILD IT

A task manager REST API with full CRUD and validation

INTERVIEWERS ASK

IoC vs DI. @Controller vs @RestController. How does auto-configuration work?

JPA and Hibernate

09

Map your objects to real tables.

WHAT TO LEARN

- Entities and mapping
- Repositories
- Lazy vs eager loading
- JPQL and native queries
- @Transactional
- Spring Data JPA
- One-to-many, many-to-many
- The N+1 problem
- Pagination and sorting
- Flyway migrations

TIME

3 weeks

BUILD IT

Move the task manager to a real DB with users, tasks and tags

INTERVIEWERS ASK

What is the N+1 problem and how do you fix it?

Security and Testing 10

Lock it down, then prove it works.

WHAT TO LEARN

- AuthN vs AuthZ
- JWT tokens
- BCrypt hashing
- Role-based access
- Mocking with Mockito
- Spring Security
- OAuth2 basics
- CORS and CSRF
- Unit tests, JUnit 5
- Testcontainers

TIME

3 weeks

BUILD IT

JWT login with USER and ADMIN roles, plus tests for each endpoint

INTERVIEWERS ASK

How does JWT work? Session vs token authentication

Docker, CI/CD and Cloud

11

Ship it like a pro.

WHAT TO LEARN

- Dockerfile
- Redis caching
- Logging with SLF4J
- Nginx basics
- Env variables, secrets
- Docker Compose
- GitHub Actions CI/CD
- Spring Actuator
- AWS: EC2, S3, RDS
- Kubernetes overview

TIME

3 weeks

BUILD IT

Containerise your API, add Redis, deploy it with a live URL

INTERVIEWERS ASK

Image vs container. How does caching cut database load?

Microservices and System Design

12

Think beyond a single server.

WHAT TO LEARN

- Monolith vs microservices
- Service discovery
- Load balancing
- Sharding, replication
- CAP theorem
- API Gateway
- Kafka or RabbitMQ
- Caching strategies
- Rate limiting
- LLD and design patterns

TIME

4 weeks

BUILD IT

Split one monolith into 3 services talking over Kafka

INTERVIEWERS ASK

Design a URL shortener. Explain the CAP theorem

SWIPE →

JAVA BACKEND ROADMAP

PROOF OF WORK

3 projects that get you shortlisted

Deploy them. Put the live links on your resume.

1 **URL shortener with analytics**
Spring Boot · PostgreSQL · Redis

2 **E-commerce backend**
JWT auth · JPA · Payments · Docker

3 **Order system on microservices**
Kafka · API Gateway · CI/CD · Cloud

RESUME RULE

One deployed project beats five tutorial clones.

SWIPE →

Comment “Java” for PDF



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